Iron Deficiency Anemia (IDA)

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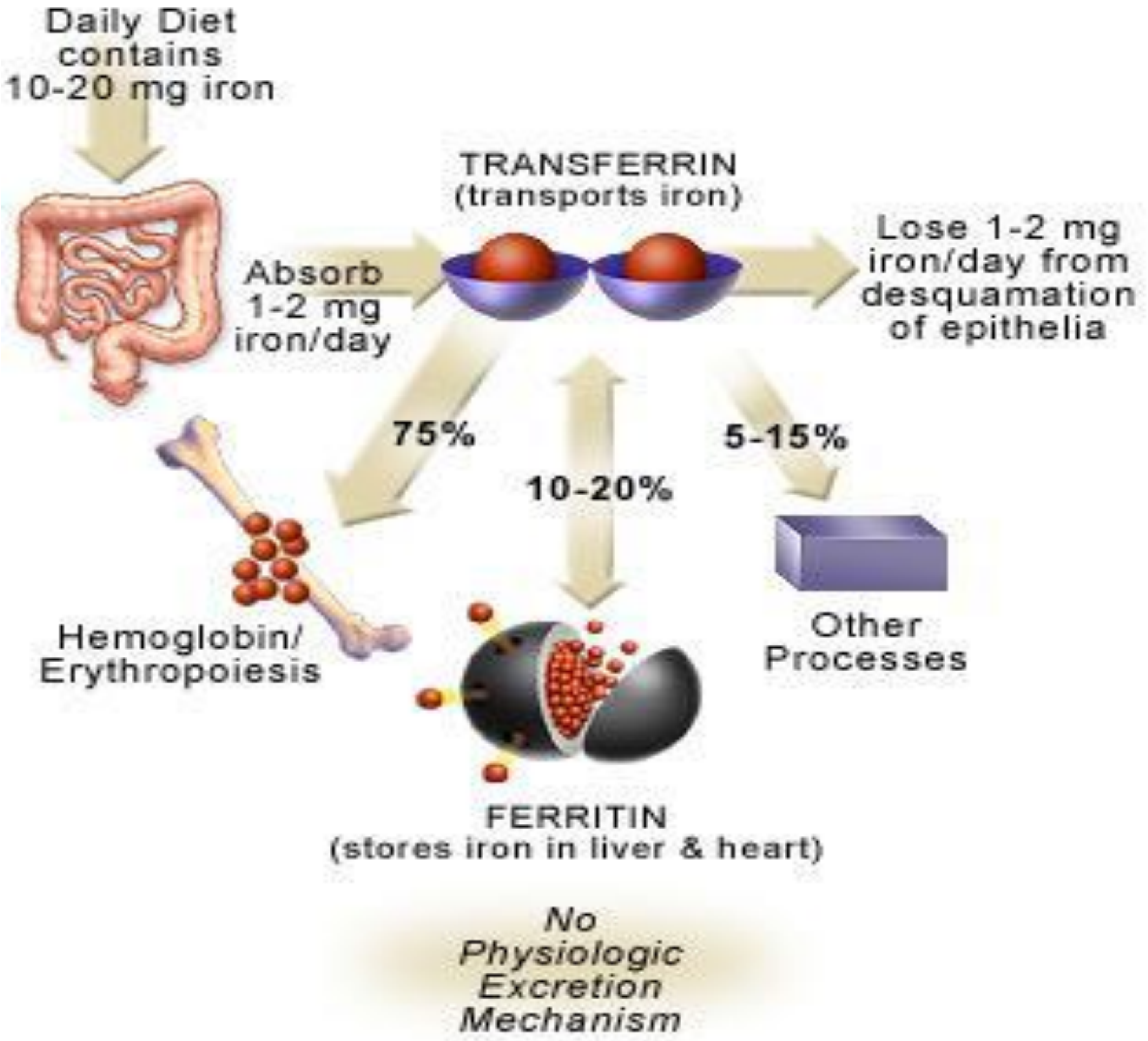
Iron

- It is both necessary for life and potentially life threatening.
- Essential element in all living cells.
- Transports oxygen (Hb).
- Stores oxygen (Mb).
- Integral part of many enzymes (Cytochrome, Catalase, Peroxidase).
- Usually bound to other molecules (not found as free).
- Absorbed in duodenum and upper part of jejunum.

- 10% of daily iron is absorbed (1–2mg/day).
- Iron balance (Quantity) is physiologically regulated by the control of iron absorption at the enterocyte.
- Iron requirements vary with the age and sex.
- Only 20% of plasma-bound iron is derived from the gut.
- Most plasma iron is derived from the breakdown of senescent (old, aging) red cells **(iron recycling)**.

Iron metabolism



Iron Deficiency Anemia (IDA)

- IDA is a common and easily treated condition that occurs when there is insufficient iron in the body (The anemia associated with inadequate iron stores).

- It is the most common type of anemia affecting 1 billion.

- In IDA, the body does not have enough iron to form hemoglobin, which means there is not enough hemoglobin to carry oxygen to the whole body.

- It is anemia of severely decreased or absent body iron stores characterized by hypochromic (not enough amount of Hb) and microcytic (more cell divisions while waiting for the cell to be filled with Hb) RBCs.

Causes of Iron deficiency

1) Inadequate intake (dietary deficiency):

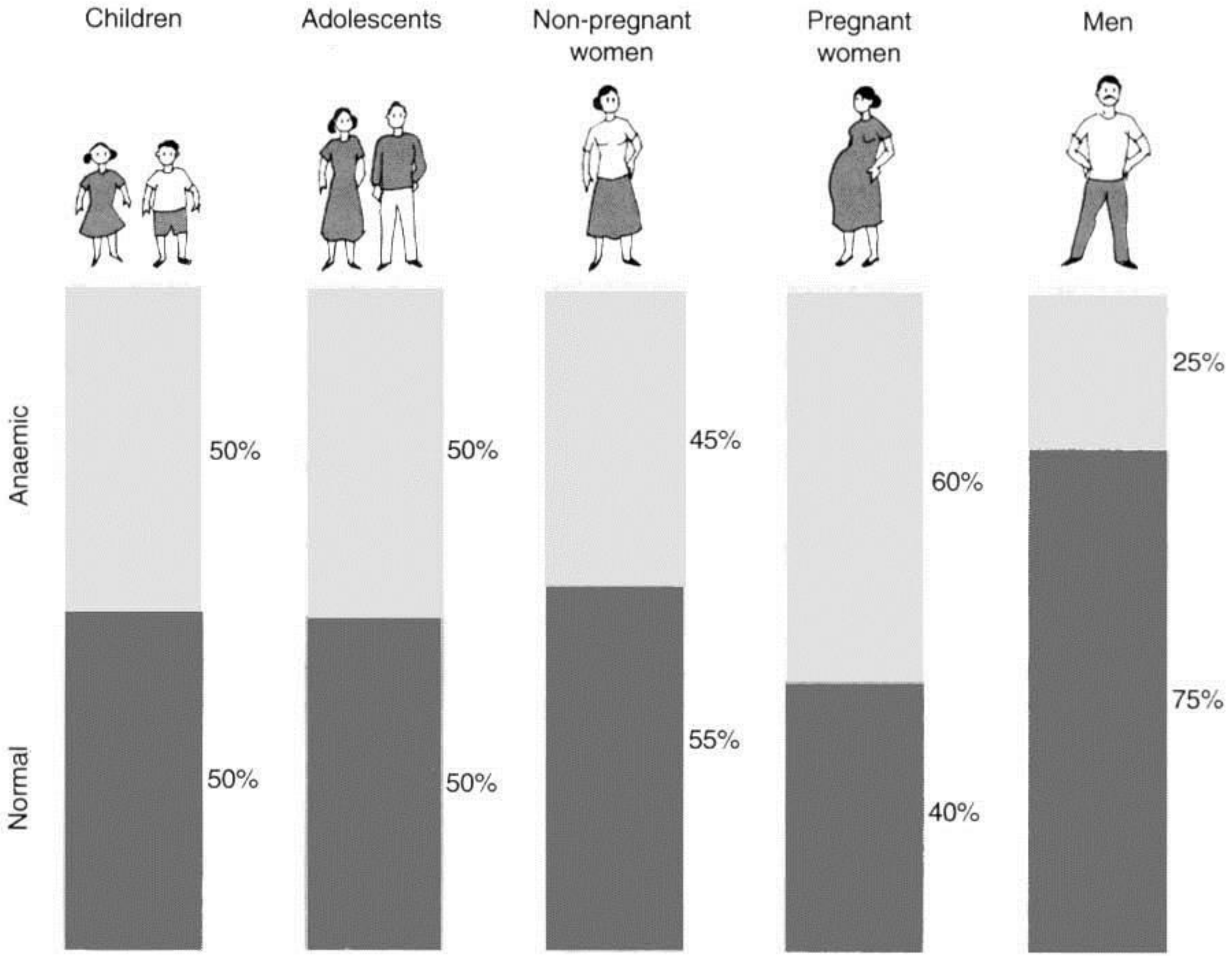
I. Meat, poultry, fish, eggs, dairy products, or iron-fortified foods (foods that have iron added) are the best sources of iron found in food.

II. The most common cause of IDA in the world

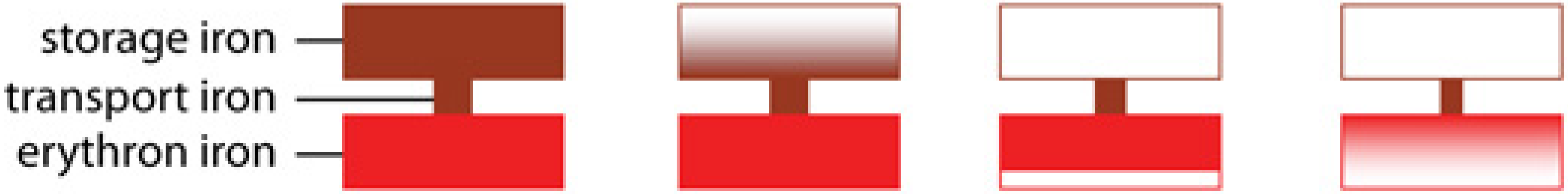
2) Increased requirements or need (intake is less than needed): during rapid growth stages (infancy, childhood, adolescence), and pregnancy and lactation (750-900mg).

3) Decreased or impaired absorption: malabsorption, gastrectomy, and others.

4) Chronic blood loss (chronic bleeding): GI bleeding (ulcers, hemorrhoids, tumors, ulcerative colitis), heavy menstrual bleeding (prolonged menorrhagia), and any condition that is associated with chronic bleeding.


Children
Adolescents
Non-pregnant women
Pregnant women
Men
Anaemic
Normal
50%
50%
50%
50%
45%
55%
60%
40%
25%
75%

	Normal	Iron depletion	Iron deficient erythropoiesis	Iron deficiency anemia



	Normal	Iron depletion	Iron deficient erythropoiesis	Iron deficiency anemia
Marrow iron	2-3+	0-1+	0	0
Serum Fe (μg/dl)	150	120	<100	<20
% saturation	40	35	<30	<20
HCT (%)	45	45	41	<40
RBC morphology	normal	normal	normal	microcytic hypochromic

Stages of Iron Status

Stage	Ferritin	Transferrin Saturation	Serum Iron	MCV	Hb	Hct
Normal	N	N	N	N	N	N
Iron Depletion	↓	N	N	N	N	N
Iron Deficient Erythropoesis	↓	↓	↓	N	N	N
Iron Deficiency Anemia	↓ ↓	↓	↓	↓	↓	↓

Major signs and symptoms of Anemia

- The major symptom of all types of anemia, including iron-deficiency anemia, is fatigue (feeling tired). Fatigue is caused by having too few red blood cells to carry oxygen to the body. This lack of oxygen can cause people to feel weak or dizzy, have a headache, or even pass out when changing position (for example, standing up).

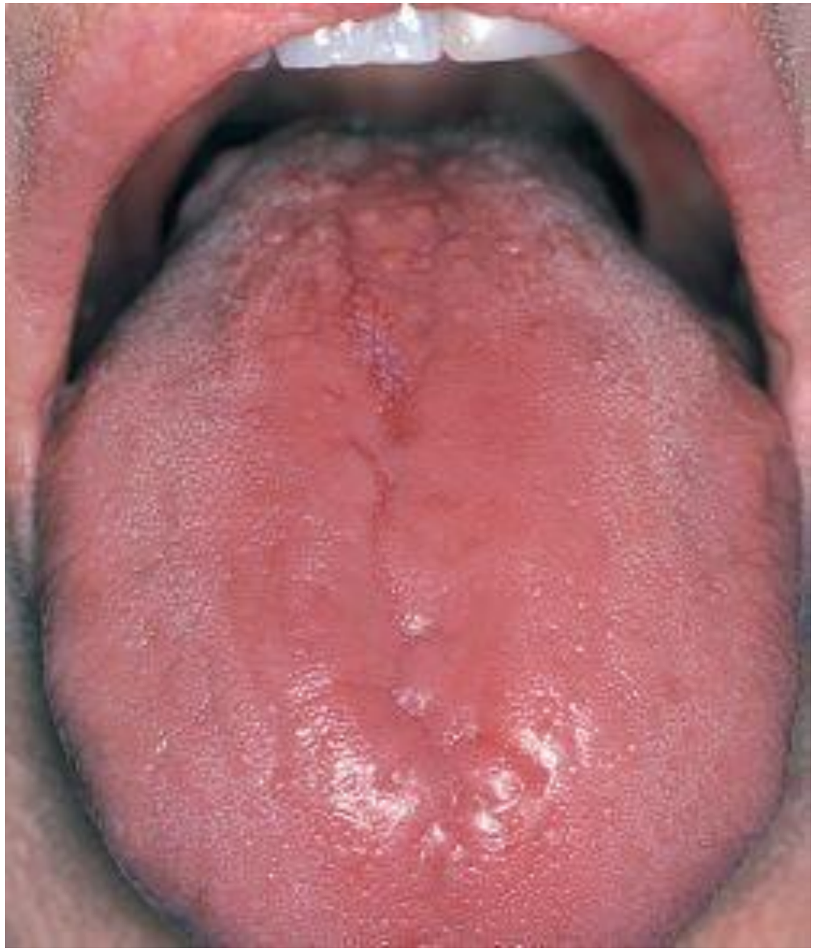
- Since the heart must work harder to move the reduced amount of oxygen, signs and symptoms may include shortness of breath and chest pain. This can lead to a fast or irregular heartbeat or a heart murmur.

- In anemia, the red blood cells don't have enough hemoglobin. Common signs of lack of hemoglobin include pale skin.

Characteristics symptoms

- Glossitis (inflammation of the tongue).
- Angular cheilosis or stomatitis (inflammation of the corners of the mouth).
- Pica (appetite for non-food substances such as ice (pagophagia), clay, starch).
- Spoon-shaped nails (Koilonychia).
- Hair loss.
- Some signs and symptoms of iron-deficiency anemia are related to its causes, such as blood loss.

Glossitis



Red, Smooth , Waxy-appearing tongue with atrophy of papillae

Concave or spoon shaped nails, koilonychia

Angular Cheilosis or Stomatitis

Ulcerations or Fissures at the Corners of the Mouth

How is iron-deficiency anemia diagnosed?

- Iron deficiency anemia is diagnosed using a person’s medical history, physical examination, and laboratory diagnostic tests.

- A doctor can use these methods to determine the verity of the anemia, its cause, and appropriate treatment.

- Mild to moderate anemia may have no signs or symptoms.

- Most patients are not diagnosed until relatively late in the progression of iron depletion.

Laboratory Tests

- **Hematological Tests :**

- ✓ CBC (screening test)
- ✓ Blood Film
- ✓ Bone Marrow Aspirate

- **Biochemical Tests (iron studies) :**

- ✓ Serum Iron
- ✓ Serum Ferritin
- ✓ Total iron-binding capacity (TIBC)
- ✓ Saturation of Transferrin

Hematological Tests

- ➢CBC: CBC do not become abnormal until late in stage 2 or early in stage 3.

❖Erythrocytes(RBCs)

- ✓Hemoglobin level ↓
- ✓Hct (PCV) ↓
- ✓RBC count ↓
- ✓MCV and MCH ↓
- ✓MCHC ↓
- ✓RDW ↑

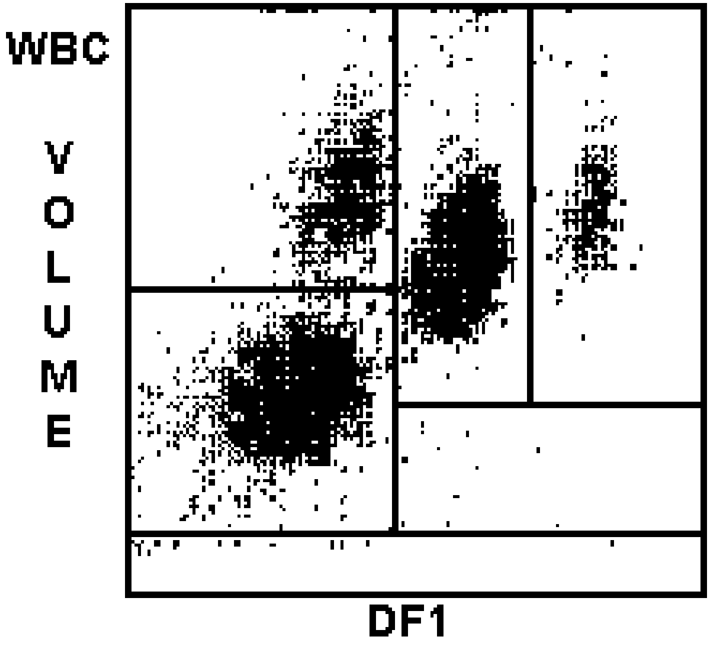
❖Leukocytes (WBCs)

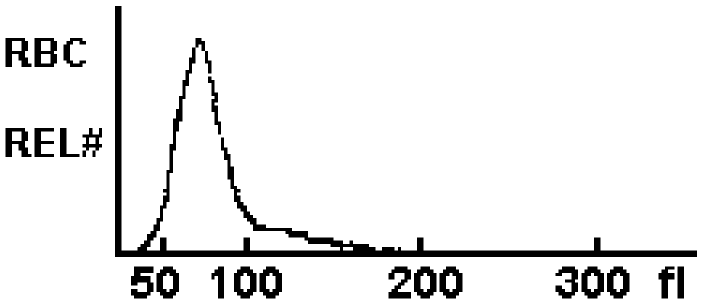
- ✓Normal

❖Platelets (Thrombocytes)

- ✓May be normal or thrombocytosis, especially if the cause is bleeding

CBC Parameters of IDA





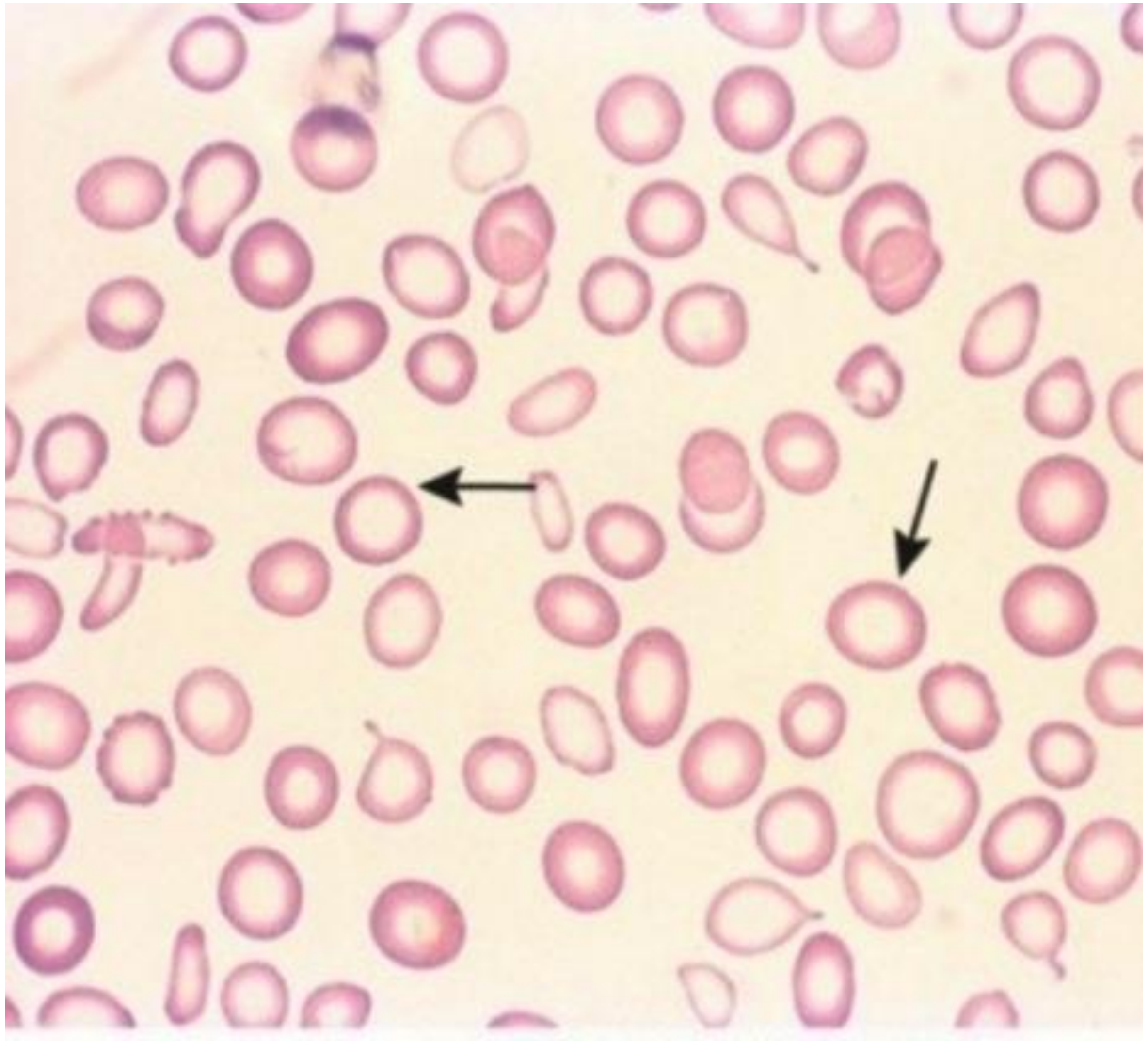
WBC	5.5		
	%		#
NE	54.7		3.0
LY	34.1		1.9
MO	7.5		0.4
EO	3.0		0.2
BA	0.7		0.0
RBC	4.28	L	
HGB	9.7	L	
HCT	29.9	L	
MCV	69.7	L	
MCH	22.6	L	
MCHC	32.4	L	
RDW	18.4	H	
PLT	331		
MPV	8.8		

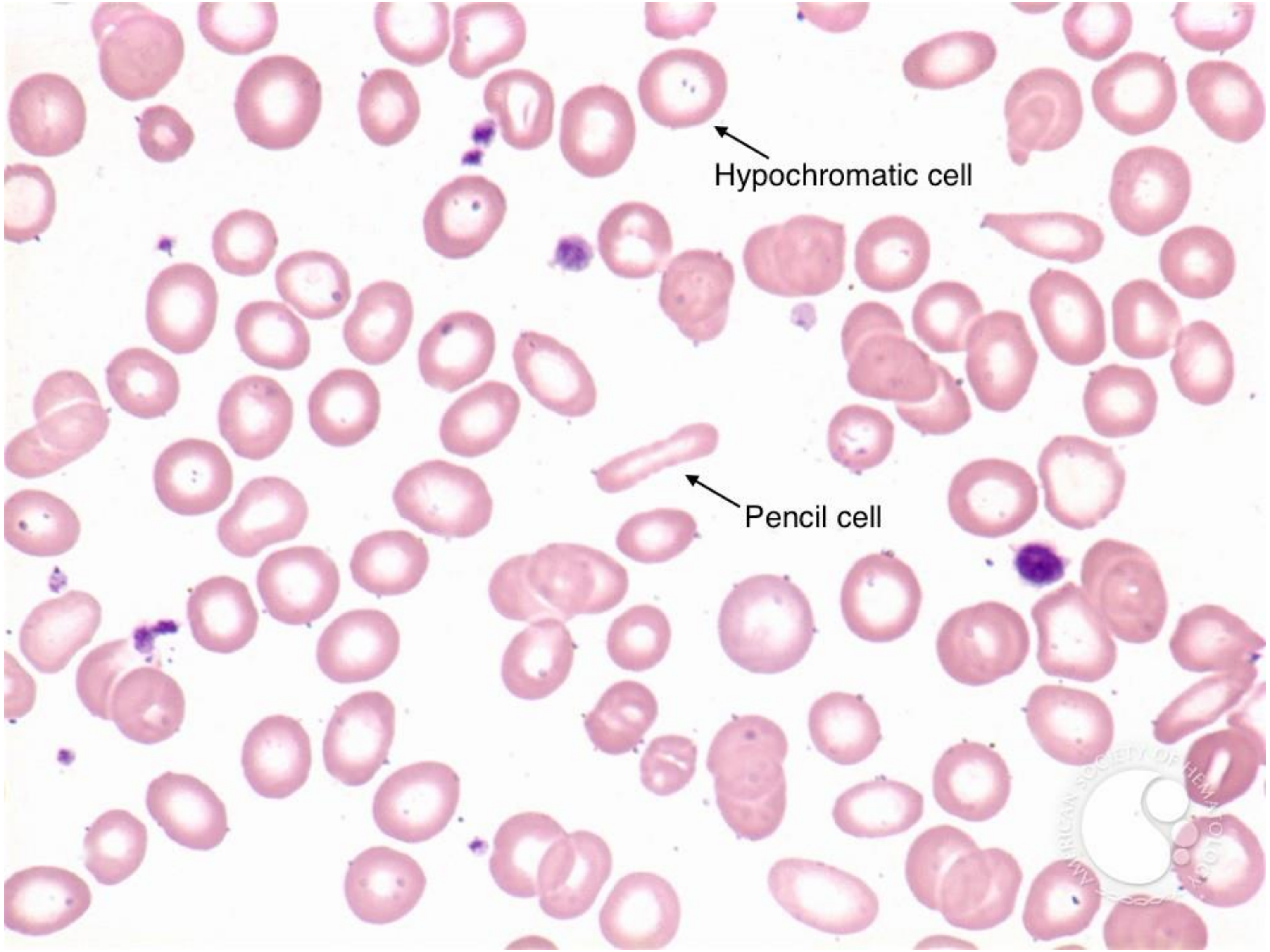
❖Blood Film:

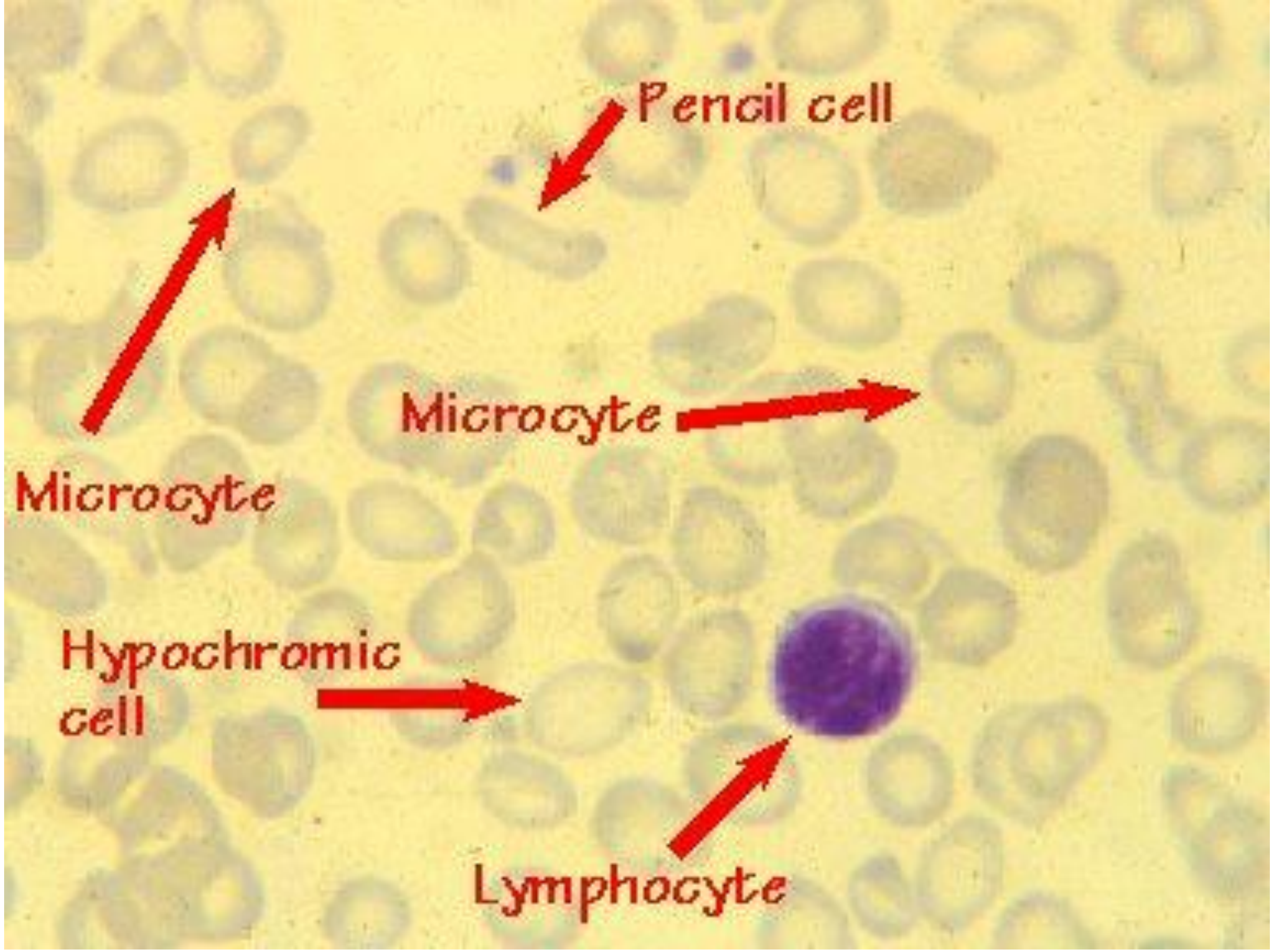
- ✓Hypochromic Microcytic RBCs.
- ✓Anisocytosis (variation in the size)
- ✓Poikilocytosis(variation in the shape)
- ✓Pencil (Cigarette) cell

- ➢Iron deficiency should be suspected when the CBC findings show a hypochromic, microcytic anemia with an elevated RDW but no consistent shape changes to the RBCs.

Hypochromic microcytic red cells

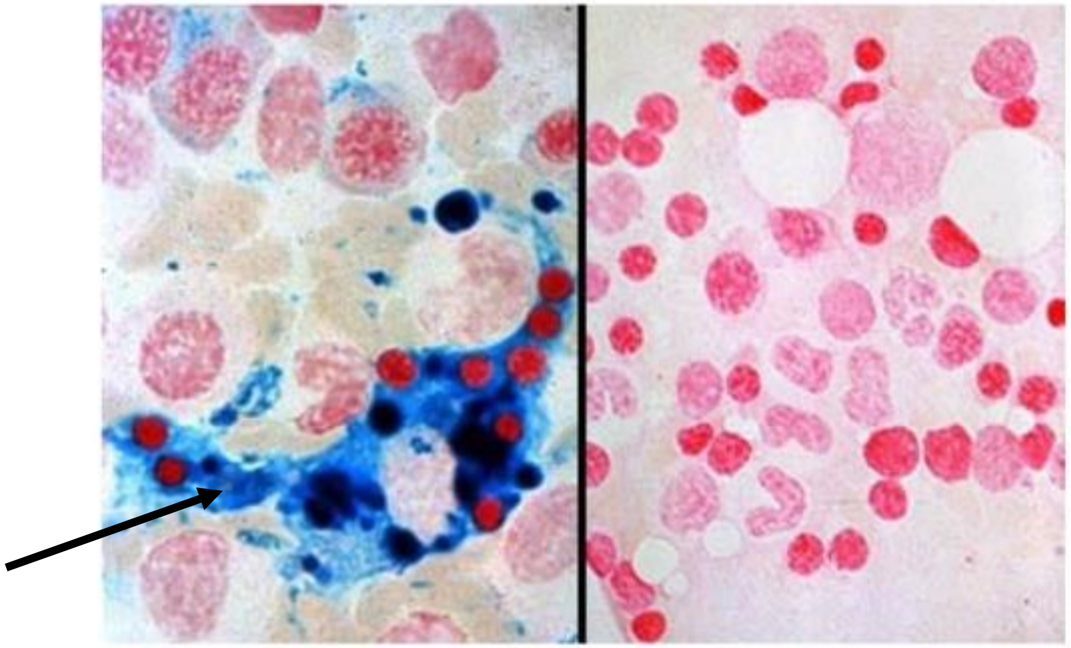



Hypochromatic cell
Pencil cell


Pencil cell
Microcyte
Microcyte
Hypochromic cell
Lymphocyte

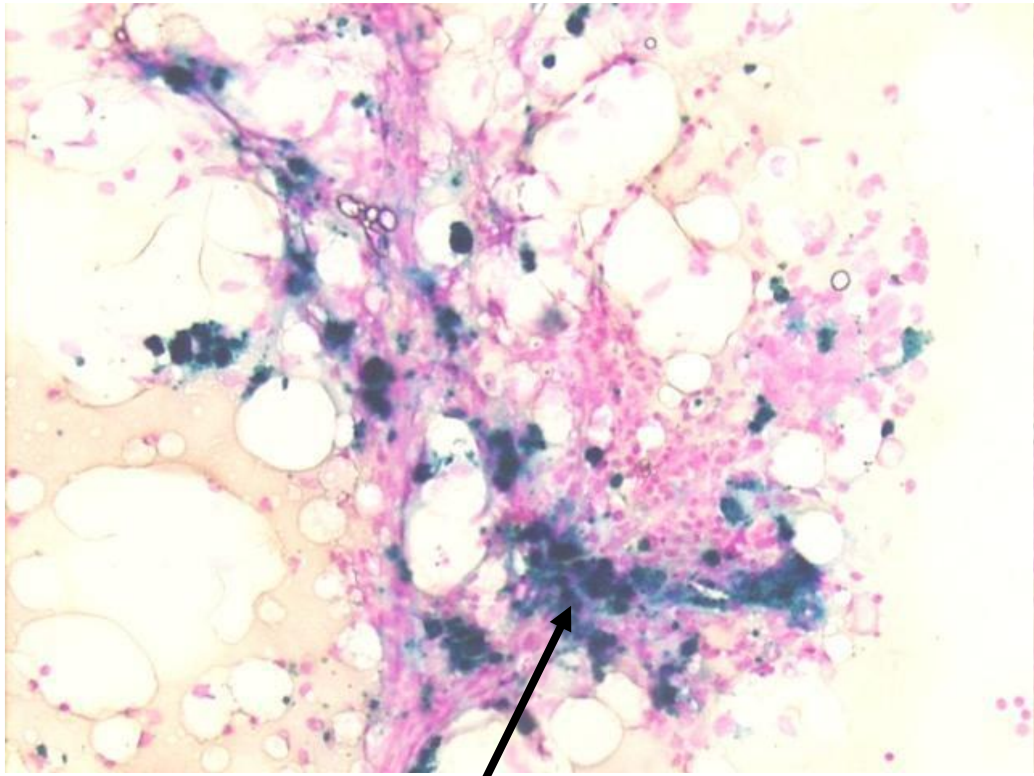
❖Bone Marrow Aspirate:

- ✓The most sensitive indicator of iron deficiency.
- ✓High cellularity (Mild to moderate erythroid hyperplasia (25-35%; N 16–18%).
- ✓Absence of stainable iron.

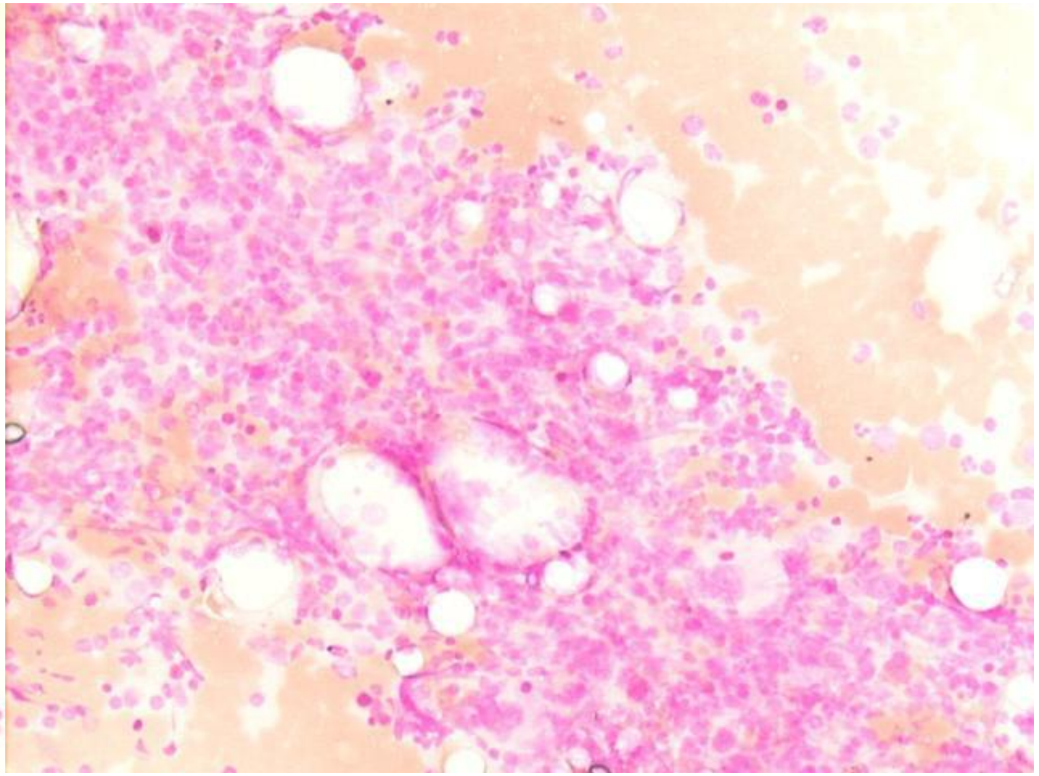


Normal Iron deficiency

Prussian Blue Stain of Bone Marrow: stains hemosiderin



Iron present



No Iron present

❖Biochemical Tests:

- ✓Iron studies remain the backbone for the diagnosis of iron deficiency.
- ✓They include assays of serum iron, total iron-binding capacity (TIBC), transferrin saturation, and serum ferritin.

❖Iron studies

- ✓Serum Iron Concentration (50-160μg/dL) ↓
- ✓*Total Iron-binding Capacity (TIBC) (250-400μg/dL)* ↑
- ✓Saturation Of Transferrin, % saturation *((Serum Iron/ TIBC) X 100)*(20–55%) ↓
- ✓Serum Ferritin Levels (40-400 ng/mL) ↓
- ✓Free erythrocyte protoporphyrin (FEP) ↑
- ✓Serum Transferrin Receptors (sTfR) ↑

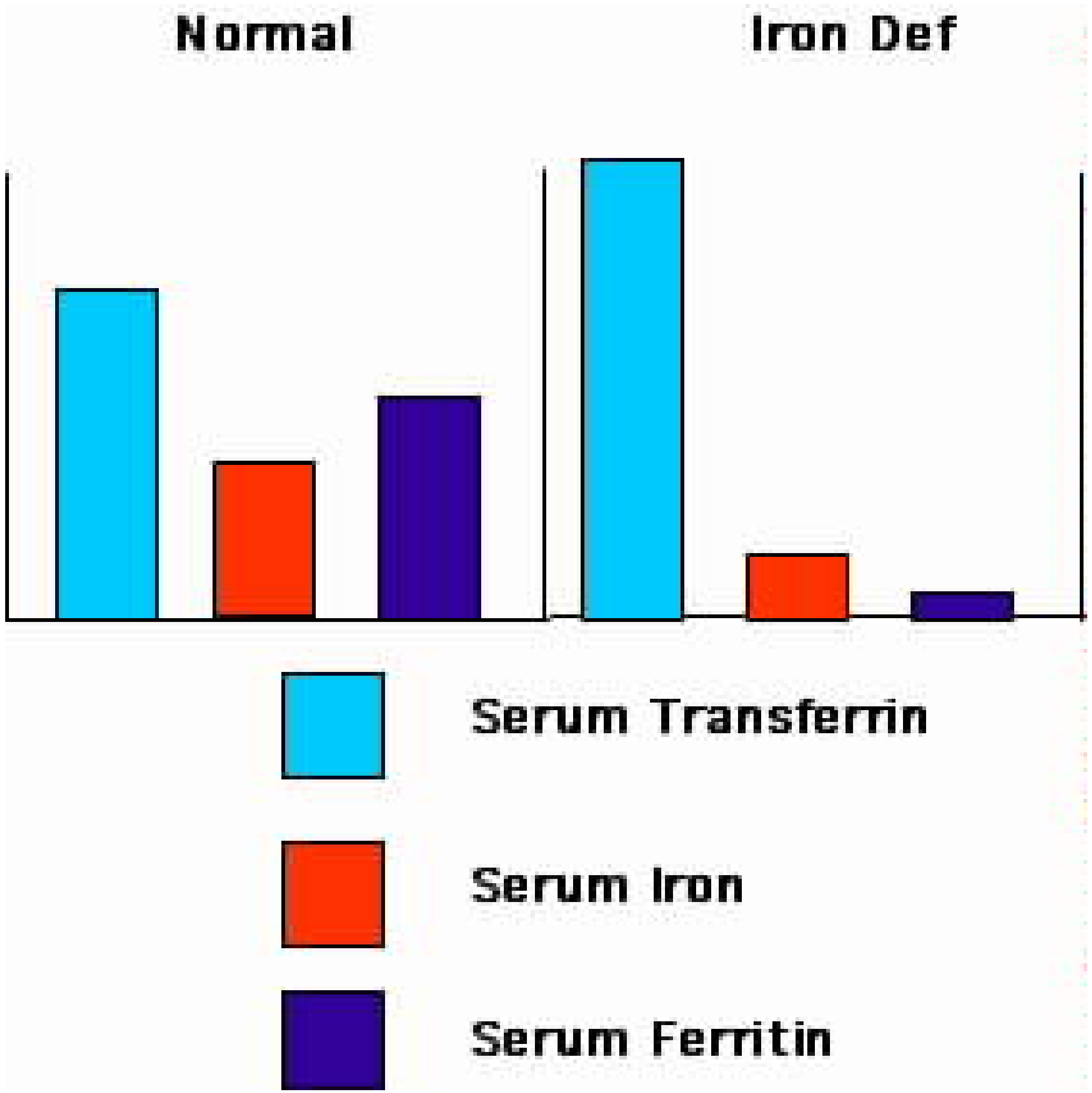
✓Serum iron measures the amount of iron bound to transferrin (transport protein) in the serum.

✓TIBC represents the total number of binding sites for iron in the plasma.

✓The percent of transferrin saturated with iron can be calculated from the total iron and the TIBC:

$$\text{Transferrin saturation (\% sat)} = \frac{\text{serum iron } (\mu\text{g/dL}) \times 100}{\text{TIBC } (\mu\text{g/dL})}$$

✓Ferritin is a measure of iron stores.


Normal
Iron Def
Serum Transferrin
Serum Iron
Serum Ferritin

Things you need to know about laboratory testing for iron status

- Iron studies include serum iron concentration, TIBC, % saturation, and ferritin.
- Ferritin is the specific test to diagnose IDA.
- Low serum ferritin is highly suggested for iron deficiency anemia.
- Normal serum ferritin does not always rule out iron deficiency.
- RDW is the first hematological parameter affected by iron deficiency.
- Hb, MCV, and MCH are the latest hematological parameters affected by iron deficiency.

IDA Treatment

- ➢ **Goals of Treatment**

- ✓ The goals of treating iron-deficiency anemia are to restore normal levels of red blood cells, hemoglobin, and iron as well as to treat the condition causing the anemia.

- ➢ **Specific Types of Treatment**

- ✓ Treatment for iron-deficiency anemia is based on the cause and the verity of the condition.
- ✓ It will include treatment to stop any bleeding, as well as changes in diet and iron supplements as needed.
- ✓ Severe anemia may require more emergency measures.

- **The best source of iron is red meat, especially beef and liver.**

  - Iron in meats is more easily absorbed by the body than iron in vegetables and other foods.
  - Chicken, turkey, fish, and shellfish are good sources of iron.
  - Other foods high in iron are eggs, peas, lentils, green leaves, and fruits.

- **Treatment for Severe and Life-Threatening Anemia**

  - Severe anemia may need to be treated with hospitalization, blood transfusions, and iron injections.

- **Oral Iron Therapy**

The optimal daily dose-200 mg of elemental iron

- Ferrous
  - Gluconate 5 tablets/ day
  - Fumarate 3 tablets/ day
  - **Sulfate 3 tablets/ day**

- Iron is absorbed more completely when the stomach is empty to maximize absorption.

- It is necessary to continue treatment for 3-6 months after the anemia is relieved to restore the iron store.

Summary

- Microcytic hypochromic anemias, those anemias that have decreased MCV and MCH below the normal range, include iron deficiency anemia (IDA).

- Most of the functional iron is found in Hb and stored in ferritin.

- IDA is the most common type of anemia and is caused by severe decreased or absent body iron stores which may be due to chronic blood loss, decreased iron intake, increased iron requirement, and decreased or impaired absorption.

- There are some people at increased risk of developing IDA: Women of childbearing age, pregnant women, people with poor diets, people who donate blood frequently, infants and children, especially those born prematurely, and vegetarians who don’t replace meat with another iron-rich food.

- IDA occurs through three stages: the depletion storage stage, the iron deficiency without anemia stage, and the IDA stage.

- RDW is the first CBC parameter that is changed in IDA.

- Hb, MCV, and MCH are the last CBC parameters changed in IDA.

- Ferritin is the specific test for IDA diagnosis.

- BM aspirate is the sensitive test for IDA.

- In IDA, the lab results are:

  ↓ Hb, MCV, MCH, MCHC, RBC count, Hct, serum iron, % saturation, and ferritin.

  ↑ RDW, TIBC

- Blood film of IDA shows hypochromic microcytic RBCs, aniso-poikilocytosis, and pencil cells.

- Characteristic signs and symptoms of IDA include pica, gastritis, angular stomatitis, and Koilonychia.